ECON 3201
Econometrics for Economics and Finance

Assignment 2
Due November 5, 2026

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Instructions

Complete your midterm exam using R and type your answers in LaTeX. Answer all questions. Please complete your assignment using Quarto. Upload both your Quarto file and pdf file to your website under new blog post.

Questions

Question 1

X and Y are discrete random variables with the following joint distribution. Use the data frame `d` to answer your questions:

```
library(kableExtra)
d <- data.frame('\\hspace{1pt}' = c(rep('Value of X',3)),
                '\\hspace{1pt}' = c(1,5,8),
                '20' = c(0.07,0.32,0.05),
                '30' = c(0.10,0.05,0.15),
                '40' = c(0.04,0.03,0.19),
                check.names = FALSE)


knitr::kable(d,
             format = "latex",
             booktabs = T,
             escape = FALSE,
             align = c("l","c","c","c","c"),
             longtable= T )%>%
  kable_styling(position = "center")%>%
```

```
  add_header_above(c(" " = 1, " " = 1, "Value of Y" = 3))%>%
  collapse_rows(columns = 1:2, valign = "top")
```

Table 1: Joint Distribution of X and Y

		Value of Y		
		20	30	40
Value of X	1	0.07	0.10	0.04
	5	0.32	0.05	0.03
	8	0.05	0.15	0.19

a) Calculate the marginal probability distribution of Y. [2 Marks]

(a) Marginal Probability Distribution of Y

The marginal probability distribution of Y is obtained by summing the joint probabilities over all values of X.

$$P(Y = 20) = 0.07 + 0.32 + 0.05 = 0.44$$

$$P(Y = 30) = 0.10 + 0.05 + 0.15 = 0.30$$

$$P(Y = 40) = 0.04 + 0.03 + 0.19 = 0.26$$

Therefore, the marginal probability distribution of Y is $P(Y = 20) = 0.44$, $P(Y = 30) = 0.30$, and $P(Y = 40) = 0.26$.

b) Calculate the mean of Y. [3 Marks] The expected value of Y is calculated using:

$$E(Y) = \sum_{y} yP(Y = y)$$

Substituting the marginal probabilities:

$$E(Y) = 20(0.44) + 30(0.30) + 40(0.26)$$

$$E(Y) = 8.8 + 9 + 10.4$$

$$\boxed{E(Y) = 28.2}$$

c) Calculate the variance of Y. [3 Marks]
The variance of Y is calculated using:

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2$$

First, calculate $E(Y^2)$:

$$E(Y^2) = \sum_y y^2 P(Y = y)$$

$$E(Y^2) = 20^2(0.44) + 30^2(0.30) + 40^2(0.26)$$

$$= 176 + 270 + 416 = 862$$

Therefore,

$$\text{Var}(Y) = 862 - (28.2)^2$$

$$= 862 - 795.24$$

$$\boxed{\text{Var}(Y) = 66.76}$$

Question 2

Using R commands, compute the following probabilities:

a) If Y is distributed $N(3, 9)$, find $Pr(Y > 1)$. [2 Marks] Given $Y \sim N(3, 9)$, the mean is $\mu = 3$ and the standard deviation is $\sigma = \sqrt{9} = 3$.

Using the standard normal distribution:

$$P(6 \leq Y \leq 8) = P\left(\frac{6-5}{\sqrt{2}} \leq Z \leq \frac{8-5}{\sqrt{2}}\right)$$

$$= P(0.7071 \leq Z \leq 2.1213)$$

$$= \Phi(2.1213) - \Phi(0.7071)$$

$$\boxed{P(6 \leq Y \leq 8) \approx 0.2228}$$

b) If Y is distributed $N(5, 2)$, find $Pr(6 \leq Y \leq 8)$. [3 Marks]
Given $Y \sim N(5, 2)$, the mean is $\mu = 5$ and the standard deviation is $\sigma = \sqrt{2}$.

Using the standard normal distribution:

$$P(6 \leq Y \leq 8) = P\left(\frac{6-5}{\sqrt{2}} \leq Z \leq \frac{8-5}{\sqrt{2}}\right)$$

$$= P(0.7071 \leq Z \leq 2.1213)$$

$$= \Phi(2.1213) - \Phi(0.7071)$$

$$\boxed{P(6 \leq Y \leq 8) \approx 0.2228}$$

Question 3

The random variable Y be a random variable with mean μ and variance σ^2. Type your answers.

a) Let Z denote the standardized version of Y. What is the functional form of Z? [1 Mark] The standardized version of Y is obtained by subtracting its mean and dividing by its standard deviation.

$$\boxed{Z = \frac{Y - \mu}{\sigma}}$$

b) Show that the expected value of of Z, $E(Z)$, in part (a) is 0. [3 Marks] Using the definition of Z:

$$E(Z) = E\left(\frac{Y - \mu}{\sigma}\right)$$

By the linearity of expectation:

$$E(Z) = \frac{1}{\sigma}[E(Y) - \mu]$$

Since $E(Y) = \mu$:

$$E(Z) = \frac{\mu - \mu}{\sigma}$$

$$\boxed{E(Z) = 0}$$

c) Show that the variance of Z, $E\left[(Z - E(Z))^2\right]$, in part (a) is 1. [3 Marks]
The variance of Z is:

$$\text{Var}(Z) = E[(Z - E(Z))^2]$$

Since $E(Z) = 0$:

$$\text{Var}(Z) = E(Z^2)$$

Substituting $Z = \frac{Y-\mu}{\sigma}$:

$$\text{Var}(Z) = E\left[\left(\frac{Y-\mu}{\sigma}\right)^2\right]$$

$$= \frac{1}{\sigma^2} E[(Y-\mu)^2]$$

Since $E[(Y-\mu)^2] = \sigma^2$:

$$= \frac{\sigma^2}{\sigma^2}$$

$$\boxed{\text{Var}(Z) = 1}$$

Question 4

Table 2 displays the joint probabilities of covid-19 vaccination status and hospitalization with covid-19.

```
library(kableExtra)
d <- data.frame('\\hspace{1pt}' = c('Hospitalized',''),
                '\\hspace{1pt}' = c('Yes','No'),
                'Yes' = c(0.007,0.666),
                'No' = c(0.075,0.252),
                check.names = FALSE)


knitr::kable(d,
             format = "latex",
             booktabs = T,
             escape = FALSE,
             align = c("l","c","c","c"),
             longtable= T)%>%
  kable_styling(position = "center")%>%
  add_header_above(c(" " = 1, " " = 1, "Vaccinated" = 2))
```

Table 2: Joint Distribution of Covid-19 Vaccination Status and Hospitalizations Due to Covid-19

		Vaccinated	
		Yes	No

Hospitalized	Yes	0.007	0.075
	No	0.666	0.252

a) Compute the probability that an individual is vaccinated conditional on being hospitalized with covid-19, i.e. Pr(Vaccinated = Yes|Hospitalized = Yes). [2 Marks] Using the conditional probability formula:

$$P(V = Yes \mid H = Yes) = \frac{P(V = Yes, H = Yes)}{P(H = Yes)}$$

From the joint distribution table:

$$P(H = Yes) = 0.007 + 0.075 = 0.082$$

Therefore,

$$P(V = Yes \mid H = Yes) = \frac{0.007}{0.082}$$

$$\boxed{P(V = Yes \mid H = Yes) = 0.0854}$$

b) Using Bayes' equation, compute the probability of being hospitalized with covid-19 conditional on being vaccinated, i.e., Pr(Hospitalized = Yes|Vaccinated = Yes). [3 Marks]

Using Bayes' theorem:

$$P(H = Yes \mid V = Yes) = \frac{P(V = Yes \mid H = Yes)P(H = Yes)}{P(V = Yes)}$$

First, calculate the marginal probability of being vaccinated:

$$P(V = Yes) = 0.007 + 0.666 = 0.673$$

From part (a):

$$P(V = Yes \mid H = Yes) = \frac{0.007}{0.082}$$

Substituting into Bayes' theorem:

$$P(H = Yes \mid V = Yes) = \frac{\left(\frac{0.007}{0.082}\right)(0.082)}{0.673}$$

$$= \frac{0.007}{0.673}$$

$$\boxed{P(H = Yes \mid V = Yes) = 0.0104}$$

Question 5

Let $Y_1, ...Y_n$ be i.i.d. draws from the population with mean μ_Y. Type your answers.

a) What is the bias of an estimator? [1 Mark] The bias of an estimator is the difference between its expected value and the true population parameter.

$$\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$$

An estimator is unbiased if its bias equals zero. b) Show that $\bar{Y}$ is an unbiased estimator of μ_Y. [5 Marks]
Given that $Y_1, \ldots, Y_n$ are independent and identically distributed (i.i.d.) random variables with population mean μ_Y:

The sample mean is defined as:

$$\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$$

Taking the expected value:

$$E(\bar{Y}) = E\left(\frac{1}{n}\sum_{i=1}^{n} Y_i\right)$$

By the linearity of expectation:

$$E(\bar{Y}) = \frac{1}{n}\sum_{i=1}^{n} E(Y_i)$$

Since the observations are identically distributed, $E(Y_i) = \mu_Y$ for all i.

$$E(\bar{Y}) = \frac{1}{n}(n\mu_Y)$$

$$\boxed{E(\bar{Y}) = \mu_Y}$$

Therefore, the bias of the sample mean is:

$$\text{Bias}(\bar{Y}) = E(\bar{Y}) - \mu_Y$$

$$= \mu_Y - \mu_Y$$

$$\boxed{\text{Bias}(\bar{Y}) = 0}$$

Hence, $\bar{Y}$ is an unbiased estimator of the population mean μ_Y.

Question 6

A researcher claims that the average height of adult men in a city is 175 cm. To test this claim, a random sample of 30 adult men is selected, and their heights are recorded. The sample has a mean height of 172 cm and a standard deviation of 8 cm.

a) State the null and alternative hypotheses for this test. [1 Mark] The bias of an estimator is the difference between its expected value and the true population parameter. The null hypothesis states that the population mean height is 175 cm, while the alternative hypothesis states that it differs from 175 cm.

$$H_0 : \mu = 175$$

$$H_1 : \mu \neq 175$$

This is a two-tailed hypothesis test.

b) Calculate the test statistic using R commands. [2 Marks] iven that $Y_1, \ldots, Y_n$ are independent and identically distributed (i.i.d.) random variables with population mean μ_Y: The test statistic is calculated using:

$$z = \frac{\bar{Y} - \mu_0}{s/\sqrt{n}}$$

Substituting the given values:

$$z = \frac{172 - 175}{8/\sqrt{30}}$$

$$\boxed{z = -2.054}$$

c) Determine the critical value(s) at the 5% significance level using the `pnorm()` command. [1 Mark] For a two-tailed test with $\alpha = 0.05$, the critical values are obtained from the standard normal distribution.

$$z_{\alpha/2} = qnorm(0.025) = -1.96$$

$$z_{1-\alpha/2} = qnorm(0.975) = 1.96$$

Therefore, the critical values are:

$$\boxed{-1.96 \text{ and } 1.96}$$

d) Based on the test statistic and critical value(s), decide whether to reject the null hypothesis. The calculated test statistic is $z = -2.054$, and the critical values are -1.96 and 1.96.

Since $-2.054 < -1.96$, the test statistic falls in the rejection region.

Therefore, we reject the null hypothesis at the 5% significance level.

There is sufficient statistical evidence to conclude that the average height of adult men in the city differs from 175 cm.

e) Construct a 95% confidence interval for the true mean height using R commands. [3 Marks] The 95% confidence interval for the population mean is:

$$CI = \bar{Y} \pm z_{0.975}\frac{s}{\sqrt{n}}$$

Substituting the values:

$$CI = 172 \pm 1.96 \left(\frac{8}{\sqrt{30}} \right)$$

$$CI = 172 \pm 2.863$$

Therefore, the 95% confidence interval is:

$$\boxed{(169.14,\ 174.86)}$$

We are 95% confident that the true population mean height lies between 169.14 cm and 174.86 cm.

Question 7

In a randomized controlled trial, 100 participants are randomly assigned to either a treatment group or a control group. The treatment group has an average improvement score of 80 with a standard deviation of 10, while the control group has an average improvement score of 75 with a standard deviation of 12.

a) Formulate the null and alternative hypotheses to test if the treatment has a significant causal effect. [2 Marks] Since the participants were randomly assigned to the treatment and control groups, we test whether the treatment has a statistically significant causal effect on the average improvement score.

The null hypothesis states that there is no difference between the population mean improvement scores of the two groups.

$$H_0 : \mu_T - \mu_C = 0$$

The alternative hypothesis states that the treatment has a significant effect.

$$H_1 : \mu_T - \mu_C \neq 0$$

This is a two-tailed hypothesis test.

b) Compute the test statistic for the difference between means using R commands. [2 Marks] Assuming the 100 participants are divided equally, each group contains 50 participants.

The test statistic for the difference between two independent sample means is:

$$z = \frac{\bar{Y}_T - \bar{Y}_C}{\sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}}}$$

Substituting the given values:

$$z = \frac{80 - 75}{\sqrt{\frac{10^2}{50} + \frac{12^2}{50}}}$$

$$z = \frac{5}{\sqrt{2 + 2.88}}$$

$$z = \frac{5}{2.2091}$$

$$\boxed{z = 2.263}$$

c) Assuming a significance level of 5%, conclude whether the treatment has a significant causal effect. [2 Marks] At the 5% significance level, the critical values for a two-tailed test are:

$$z_{critical} = \pm 1.96$$

The calculated test statistic is:

$$z = 2.263$$

Since $|2.263| > 1.96$, the test statistic falls in the rejection region.

Therefore, we reject the null hypothesis at the 5% significance level.

There is sufficient evidence to conclude that the treatment group has a higher average improvement score than the control group.